

Table of Contents

1. Abstract
2. Introduction
3. Problem Statement
4. Objectives
5. Theoretical Background
 - 5.1 Atmospheric Image Degradation
 - 5.2 Effects of Mist on Image Quality
 - 5.3 Histogram Characteristics of Misty Images
6. Literature Review
7. Proposed Methodology
8. Detailed Methodology and Analysis
 - 8.1 Image Acquisition
 - 8.2 RGB to LAB Color Space Conversion
 - 8.3 Luminance Normalization
 - 8.4 CLAHE Enhancement
 - 8.5 Gamma Correction
 - 8.6 Morphological Top-Hat Filtering
 - 8.7 Gaussian Spatial Filtering
 - 8.8 Contrast Stretching
 - 8.9 LAB to RGB Reconstruction
 - 8.10 Final Gaussian Smoothing
 - 8.11 Unsharp Masking
9. Flowchart of Proposed System
10. Algorithm
11. Parameter Selection and Justification
12. Results and Discussion
13. Histogram Analysis
14. Performance Evaluation
15. Strengths of the Proposed Method
16. Limitations
17. Applications
18. Future Work
19. Conclusion
20. MATLAB Code
21. Output Images
22. References

1. Abstract

Outdoor imaging systems often suffer from visibility degradation due to atmospheric phenomena such as mist, haze, and fog. Mist introduces a whitish veil over captured scenes because of light scattering caused by tiny water particles suspended in the atmosphere. As a result, images experience reduced contrast, faded colors, blurred edges, and loss of important visual information. These degradations negatively affect both human visual interpretation and computer vision systems.

This project presents a MATLAB-based mist removal framework using classical image processing techniques. The proposed system enhances mist-degraded images through a sequence of operations including LAB color space transformation, adaptive histogram equalization, gamma correction, morphological filtering, Gaussian spatial filtering, contrast stretching, and unsharp masking. The methodology focuses on improving image visibility while preserving natural colors and structural details.

The proposed method effectively restores scene clarity, enhances local contrast, sharpens blurred edges, and improves overall perceptual quality without requiring computationally expensive deep learning models. Experimental results demonstrate that the system successfully reduces mist effects and produces visually clearer images suitable for surveillance, outdoor monitoring, intelligent transportation systems, and computer vision preprocessing applications.

Keywords: Mist Removal, Image Enhancement, CLAHE, Morphological Filtering, Contrast Stretching, MATLAB, Digital Image Processing.

2. Introduction

Digital images captured in outdoor environments are frequently affected by atmospheric conditions such as mist, haze, smoke, fog, and dust. Among these degradations, mist is one of the most common environmental factors that significantly reduces image quality. Mist occurs due to the presence of tiny water droplets suspended in the atmosphere, which scatter incoming light before it reaches the camera sensor. This scattering process introduces a translucent veil over the scene and causes visibility reduction.

Mist degradation creates several visual problems in an image. The contrast between bright and dark regions decreases, edges become blurred, colors appear faded, and fine textures become less visible. These effects not only reduce visual quality for human observers but also negatively affect the performance of computer vision systems such as object detection, autonomous driving systems, surveillance systems, robotics, remote sensing, and traffic monitoring systems.

Image enhancement techniques are commonly used to restore degraded images and improve their visual appearance. Traditional mist removal approaches rely on image processing algorithms that enhance luminance, restore contrast, and suppress

atmospheric artifacts. Compared to deep learning approaches, classical image processing techniques require lower computational resources, are easier to implement, and provide faster execution on standard hardware.

This project focuses on developing a MATLAB-based mist removal framework using classical digital image processing techniques. The proposed method combines multiple enhancement operations to compensate for the effects of atmospheric scattering. The enhancement pipeline includes adaptive contrast enhancement, nonlinear brightness correction, morphological processing, spatial-domain filtering, contrast stretching, and image sharpening.

The developed system aims to produce visually clear images while preserving natural color appearance and important edge details. The methodology is computationally efficient and suitable for real-world image enhancement applications.

3. Problem Statement

Mist significantly degrades image quality by reducing contrast, suppressing visibility, desaturating colors, and introducing a grayish or whitish atmospheric veil across the scene. Due to atmospheric scattering, light reflected from scene objects becomes partially scattered before reaching the imaging sensor. This effect reduces local intensity variations and causes distant objects to appear unclear and faded.

In mist-degraded images:

- Important scene details become hidden.
- Edge sharpness decreases.
- The dynamic range of intensity values becomes compressed.
- Color information becomes weak and unnatural.
- Overall scene visibility becomes poor.

Such degradations negatively affect image interpretation and computer vision analysis. Therefore, an effective image enhancement framework is required to restore visibility and improve image quality.

The primary goal of this project is to develop a MATLAB-based mist removal system using classical image processing techniques to:

- Restore image contrast and visibility.
- Preserve natural colors and textures.
- Enhance edge details blurred by mist.
- Improve overall perceptual quality.
- Produce computationally efficient image enhancement.

4. Objectives

The objectives of this project are:

- To study the visual effects of mist on digital images.
- To analyze atmospheric degradation and visibility reduction.
- To implement a MATLAB-based mist removal pipeline.
- To enhance image contrast using adaptive histogram equalization.
- To improve brightness distribution using gamma correction.
- To suppress mist-related illumination using morphological filtering.
- To reduce enhancement noise using Gaussian filtering.
- To sharpen blurred image details using unsharp masking.
- To evaluate the visual performance of the enhancement pipeline.
- To produce a computationally efficient mist removal framework.

5. Theoretical Background

5.1 Atmospheric Image Degradation

Atmospheric particles present in the environment interact with light traveling from scene objects toward the camera sensor. During this process, a portion of the reflected light becomes scattered in different directions. This phenomenon is called atmospheric scattering.

Mist mainly consists of small water droplets suspended in air. These droplets scatter incoming light and introduce an additional atmospheric component known as air light. Air light creates a whitish veil over the image and reduces scene visibility.

The degradation process can be mathematically represented as:

$$J(x)=I(x)t(x)+A(1-t(x))$$

where:

- $J(x)$ represents the observed misty image.
- $I(x)$ represents the original scene radiance.
- $t(x)$ represents transmission.
- A represents atmospheric light.

As transmission decreases, the visibility of scene objects also decreases.

5.2 Effects of Mist on Image Quality

Mist affects digital images in several ways.

5.2.1 Reduction in Contrast

Contrast represents the difference between bright and dark intensity regions in an image. Mist compresses the intensity distribution and reduces local luminance variation. As a result, objects appear flat and less distinguishable.

5.2.2 Color Desaturation

Mist weakens color information by introducing atmospheric whiteness into the scene. Colors lose saturation and become pale or washed out.

5.2.3 Edge Blurring

Fine image details become difficult to observe because mist reduces high-frequency intensity variations. Edges become blurred and structural details become weak.

5.2.4 Reduced Visibility

Distant objects experience stronger atmospheric attenuation compared to nearby objects. Therefore, distant scene regions appear unclear and heavily faded.

5.3 Histogram Characteristics of Misty Images

The histogram of a mist-degraded image usually becomes narrow and concentrated around mid-intensity values. This occurs because mist reduces the dynamic range of intensity values.

Characteristics of misty image histograms include:

- Narrow intensity distribution.
- Reduced contrast.
- Weak dark and bright regions.
- Concentration around central intensity levels.

Image enhancement techniques aim to redistribute these intensity values across the full dynamic range.

6. Literature Review

Numerous techniques have been proposed for atmospheric image enhancement and mist removal. Existing methods can generally be categorized into classical image processing approaches and deep learning-based approaches.

Classical image enhancement techniques include histogram equalization, contrast stretching, dark channel prior, and morphological filtering methods. These approaches are computationally efficient and relatively easy to implement.

Adaptive Histogram Equalization (AHE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) are commonly used for enhancing local image contrast. CLAHE prevents excessive amplification of noise by limiting histogram clipping.

Morphological operations are also effective for removing uneven illumination components caused by atmospheric effects. Top-hat filtering can isolate slowly varying background illumination and improve local visibility.

Deep learning techniques have recently gained popularity for haze and mist removal. Convolutional neural networks can estimate atmospheric transmission maps and reconstruct clear images. However, these approaches require large training datasets, high computational resources, and powerful hardware.

In this project, classical image processing techniques were selected because of their simplicity, efficiency, interpretability, and suitability for real-time enhancement systems.

7. Proposed Methodology

The proposed methodology combines multiple enhancement techniques to restore mist-degraded images. Each stage of the enhancement pipeline targets a specific degradation effect introduced by mist.

The complete enhancement pipeline consists of the following steps:

- Image acquisition
- RGB to LAB conversion
- Luminance normalization
- CLAHE enhancement
- Gamma correction
- Morphological filtering
- Gaussian spatial filtering
- Contrast stretching
- LAB to RGB reconstruction
- Final smoothing and sharpening

8. Detailed Methodology and Analysis

8.1 Image Acquisition

The enhancement process begins by reading the mist-degraded RGB image using MATLAB. The image is converted into double precision format to allow accurate mathematical processing during enhancement operations.

The original image contains reduced visibility, faded colors, and poor contrast due to atmospheric mist.

8.2 RGB to LAB Color Space Conversion

The RGB image is converted into LAB color space using the `rgb2lab` function.

LAB color space separates luminance information from color information:

- L channel represents brightness.
- a channel represents green-red color information.
- b channel represents blue-yellow color information.

This separation is extremely useful because mist mainly affects luminance and visibility rather than color components directly. Therefore, enhancement can be applied selectively to the luminance channel without heavily distorting image colors.

Compared to direct RGB enhancement, LAB-based processing preserves more natural color appearance.

8.3 Luminance Normalization

The L channel is normalized using the `mat2gray` function.

Normalization converts luminance values into the range $[0, 1]$. This standardization improves numerical stability and prepares the image for adaptive contrast enhancement.

Without normalization, enhancement operations may produce inconsistent brightness scaling.

8.4 CLAHE Enhancement

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to improve local contrast.

Unlike global histogram equalization, CLAHE divides the image into small contextual regions called tiles. Histogram equalization is performed independently within each tile. This allows localized enhancement of mist-affected regions.

The clip limit parameter prevents excessive amplification of noise and avoids over-enhancement in homogeneous regions.

The selected parameters are:

- ClipLimit = 0.02
- NumTiles = [8 8]

These values provide balanced enhancement while preserving natural appearance.

CLAHE effectively restores local intensity variation suppressed by atmospheric scattering.

8.5 Gamma Correction

Gamma correction is applied after CLAHE to improve brightness distribution.

Gamma correction performs nonlinear intensity transformation according to:

$$I_{out} = I_{in}^{\gamma}$$

where gamma controls brightness behavior.

A gamma value less than 1 brightens darker regions and improves visibility in mist-obscured areas.

In this project:

$$\text{Gamma} = 0.68$$

This value enhances mid-tone luminance while avoiding excessive brightness.

Gamma correction compensates for illumination attenuation caused by atmospheric mist.

8.6 Morphological Top-Hat Filtering

Morphological filtering is used to suppress smooth illumination components introduced by mist.

Top-hat transformation extracts uneven background illumination according to:

$$\text{TopHat} = \text{Image} - \text{Opening}(\text{Image})$$

Mist behaves like a low-frequency illumination layer spread across the image. Morphological opening estimates this smooth background component.

A disk-shaped structuring element is used:

Structuring Element = disk(10)

The selected disk size captures large-scale atmospheric illumination variations without affecting important structural details.

Morphological processing significantly improves local visibility and removes unwanted atmospheric brightness.

8.7 Gaussian Spatial Filtering

After enhancement operations, small noise artifacts may appear in the image.

Gaussian filtering is applied to smooth residual noise while preserving important structures.

Gaussian filtering performs weighted averaging using a Gaussian kernel. Neighboring pixels closer to the center contribute more strongly to the filtering process.

The Gaussian function is represented as:

$$G(x,y)=1/(2\pi\sigma^2)e^{-(x^2+y^2)/(2*\sigma^2)}$$

The selected parameter is:

Sigma = 0.95

This value provides moderate smoothing without excessive blur.

8.8 Contrast Stretching

Contrast stretching is applied to expand the intensity range and improve overall visibility.

The stretchlim function estimates lower and upper intensity boundaries, while imadjust redistributes intensity values across the full dynamic range.

Selected parameters:

- Lower Limit = 0.001
- Upper Limit = 0.995

These values improve contrast while avoiding saturation artifacts.

Contrast stretching restores intensity distribution compressed due to mist.

8.9 LAB to RGB Reconstruction

After luminance enhancement, the processed L channel is recombined with the original a and b color channels.

The image is then converted back to RGB format using lab2rgb.

This reconstruction step restores natural color appearance while preserving enhanced visibility.

8.10 Final Gaussian Smoothing

A second Gaussian smoothing operation is applied to suppress small enhancement artifacts generated during reconstruction.

This step improves visual smoothness and creates more natural image appearance.

8.11 Unsharp Masking

Unsharp masking is used to restore edges weakened by atmospheric scattering.

Sharpening is achieved by enhancing high-frequency components of the image.

The sharpening operation can be represented as:

$$I_{\text{sharp}} = I + k(I - I_{\text{blur}})$$

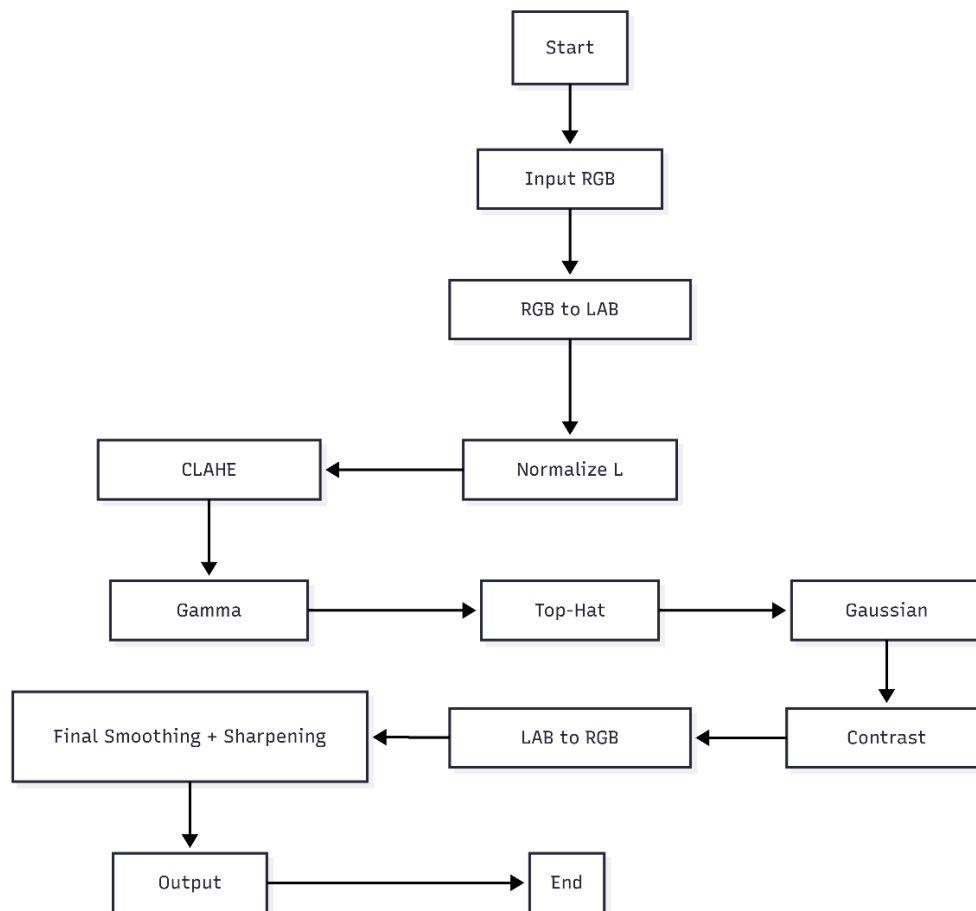
where:

- I represents the original image.
- I_{blur} represents blurred image.
- k controls sharpening strength.

A mild sharpening filter is selected to avoid artificial appearance.

This operation improves edge visibility and restores fine structural details.

9. Flowchart of Proposed System



10. Algorithm

Algorithm for Mist Removal

Step 1: Read mist-degraded RGB image.

Step 2: Convert RGB image into LAB color space.

Step 3: Extract luminance channel.

Step 4: Normalize luminance values.

Step 5: Apply CLAHE enhancement.

Step 6: Perform gamma correction.

Step 7: Apply morphological top-hat filtering.

Step 8: Smooth residual noise using Gaussian filtering.

Step 9: Apply contrast stretching.

Step 10: Reconstruct RGB image.

Step 11: Apply final Gaussian smoothing.

Step 12: Perform unsharp masking.

Step 13: Display and save enhanced image.

11. Parameter Selection and Justification

Parameter	Value	Justification
CLAHE ClipLimit	0.02	Prevents excessive noise amplification
CLAHE Tiles	[8 8]	Provides balanced local enhancement
Gamma Value	0.68	Enhances mist-darkened regions
Structuring Element	disk(10)	Captures smooth mist illumination
Gaussian Sigma	0.95	Removes noise while preserving details
Stretch Limits	0.001–0.995	Expands dynamic intensity range
Unsharp Sigma	0.5	Mild sharpening without artifacts

12. Results and Discussion

The proposed enhancement pipeline successfully improves the visibility of mist-degraded images. Significant improvements are observed in image contrast, edge sharpness, and color quality.

The original misty image contains faded colors, weak textures, reduced visibility, and compressed intensity distribution. After enhancement, scene details become more distinguishable and structural information becomes clearer.

CLAHE enhancement effectively restores local contrast in regions affected by atmospheric scattering. Gamma correction improves brightness distribution and reveals hidden details in darker image areas.

Morphological filtering suppresses smooth atmospheric illumination components and improves local visibility. Gaussian filtering removes enhancement noise while maintaining image structure.

Contrast stretching expands the dynamic range and produces stronger intensity separation between objects. Final sharpening improves edge visibility and restores structural details weakened by mist.

The resulting image appears significantly clearer while maintaining natural color appearance.

13. Histogram Analysis

Histogram analysis demonstrates the effectiveness of the proposed enhancement pipeline.

13.1 Original Image Histogram

The histogram of the original misty image is concentrated around middle intensity values. This narrow distribution indicates reduced contrast and compressed dynamic range.

Characteristics include:

- Weak dark regions
- Weak bright regions
- Low intensity variation
- Poor contrast distribution

13.2 Enhanced Image Histogram

After enhancement, the histogram becomes more widely distributed across the intensity range.

Characteristics include:

- Improved dynamic range
- Stronger contrast distribution
- Better separation of dark and bright regions
- Enhanced visibility of image structures

This histogram expansion confirms successful restoration of image contrast.

14. Performance Evaluation

14.1 Visual Performance

Metric	Original Image	Enhanced Image
Contrast	Low	High
Visibility	Poor	Clear
Color Quality	Dull	Restored
Edge Sharpness	Blurred	Sharpened
Texture Visibility	Weak	Improved

14.2 Computational Efficiency

The proposed method uses classical spatial-domain image processing operations. Therefore, the computational complexity remains relatively low compared to deep learning approaches.

Advantages include:

- Fast execution
- Low memory requirements
- No training data required
- Suitable for real-time preprocessing
- Easy MATLAB implementation

The enhancement pipeline is computationally efficient and practical for lightweight imaging systems.

15. Strengths of the Proposed Method

The proposed method provides several advantages:

- Effective removal of mist-related degradation.
- Preservation of natural image colors.
- Improved edge and texture visibility.
- Computationally efficient implementation.
- Easy integration with MATLAB systems.
- No requirement for training datasets.
- Suitable for real-time enhancement applications.

16. Limitations

Despite successful enhancement performance, several limitations exist. Extremely dense mist conditions may not be fully corrected. Parameter tuning may be required for different image conditions. Excessive enhancement can occasionally produce unnatural brightness. Fine textures may experience slight smoothing. The method does not estimate physical atmospheric depth.

17. Applications

The proposed mist removal system can be used in various real-world applications including:

- Surveillance systems
- Traffic monitoring
- Autonomous vehicles
- Remote sensing
- Drone imaging
- Medical imaging preprocessing
- Outdoor photography enhancement
- Robotics and navigation systems

18. Future Work

Several improvements can be explored in future work.

- Adaptive parameter optimization.
- Integration with deep learning models.
- Real-time FPGA implementation.
- Automatic mist density estimation.
- Multi-scale enhancement approaches.
- Video-based mist removal systems.
- Hybrid spatial-frequency domain enhancement.

Future research can improve robustness under severe atmospheric conditions.

19. Conclusion

This project successfully presents a MATLAB-based mist removal framework using classical digital image processing techniques. The proposed enhancement pipeline combines LAB color space processing, CLAHE enhancement, gamma correction, morphological filtering, Gaussian smoothing, contrast stretching, and unsharp masking to restore visibility in mist-degraded images.

The methodology effectively compensates for atmospheric scattering effects and significantly improves image clarity, local contrast, edge sharpness, and perceptual quality. Experimental analysis demonstrates that the enhanced images exhibit stronger visibility, restored color appearance, and improved structural detail preservation. Compared to computationally expensive deep learning methods, the proposed approach is lightweight, interpretable, and computationally efficient. Therefore, it is suitable for real-time image enhancement and practical computer vision preprocessing applications.